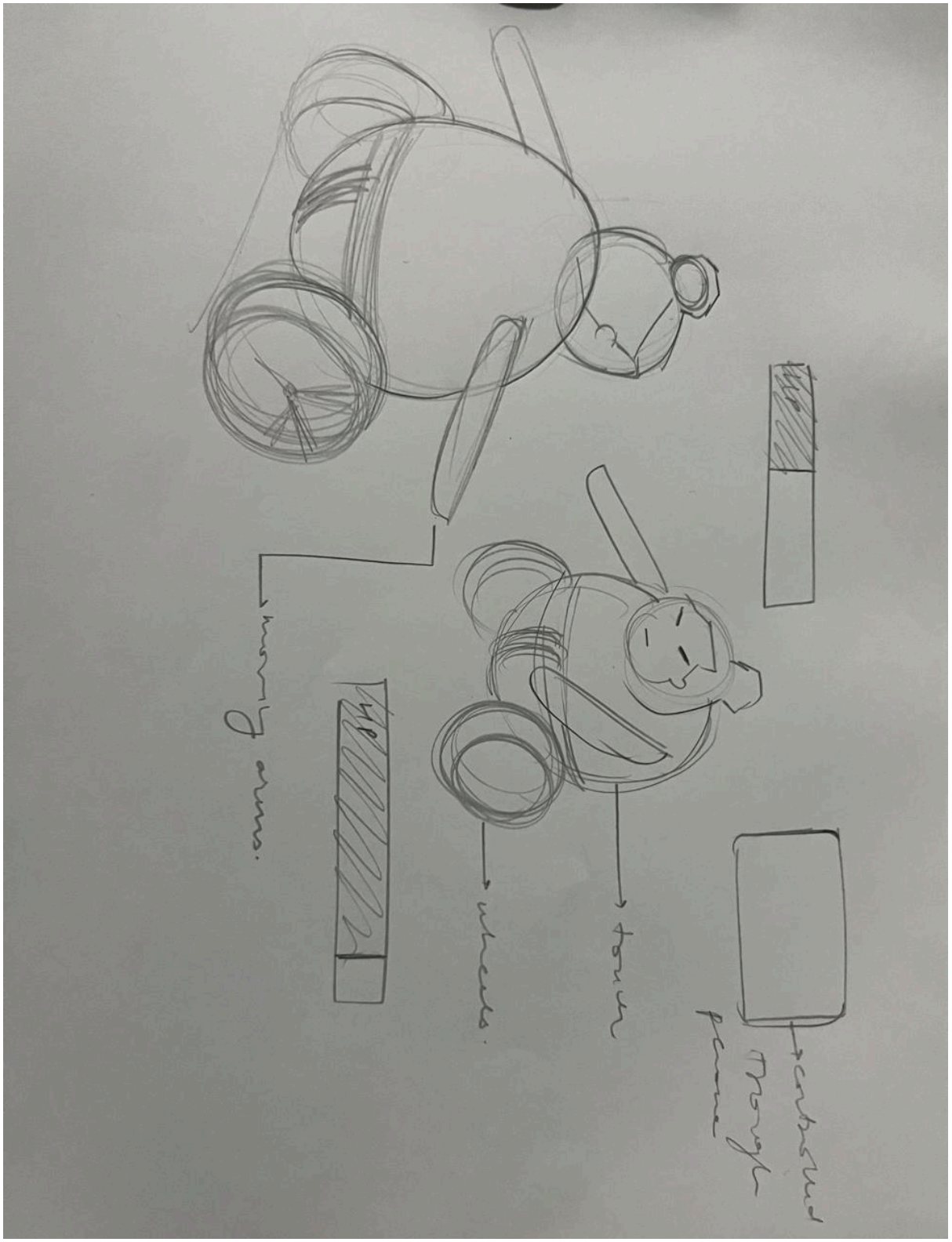
ODT Summative 2

Idea Brief :

This project implements a **two-player robotic combat system** consisting of autonomous **sumo-style bots**, each controlled wirelessly via a mobile device. Each bot operates as an independent embedded system built around an **ESP32 microcontroller**, enabling real-time communication, control, and feedback.

The bots engage within a defined arena (ring), where the objective is to **deplete the opponent's health points (HP)** through physical contact or force the opponent out of the boundary. Contact between bots is detected using a **capacitive touch sensing mechanism**, triggering a decrement in the opponent's HP. A bot is eliminated when its HP reaches zero or when it exits the arena.

Basic Idea Sketch :